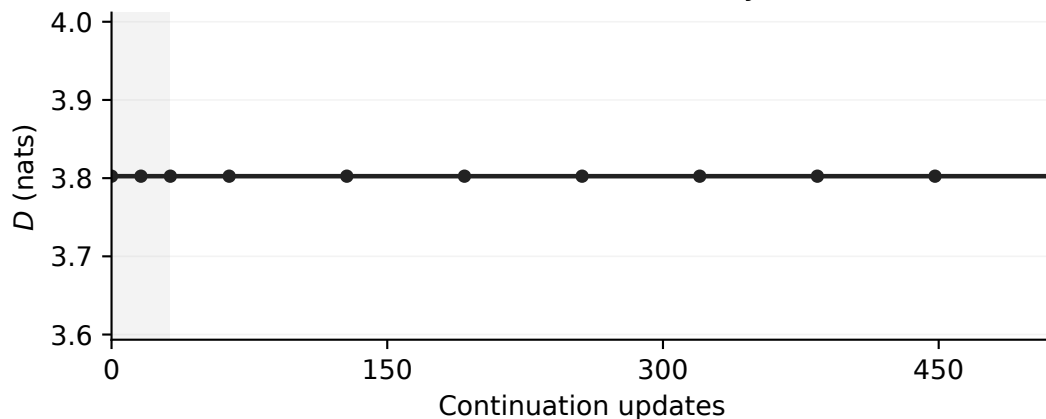
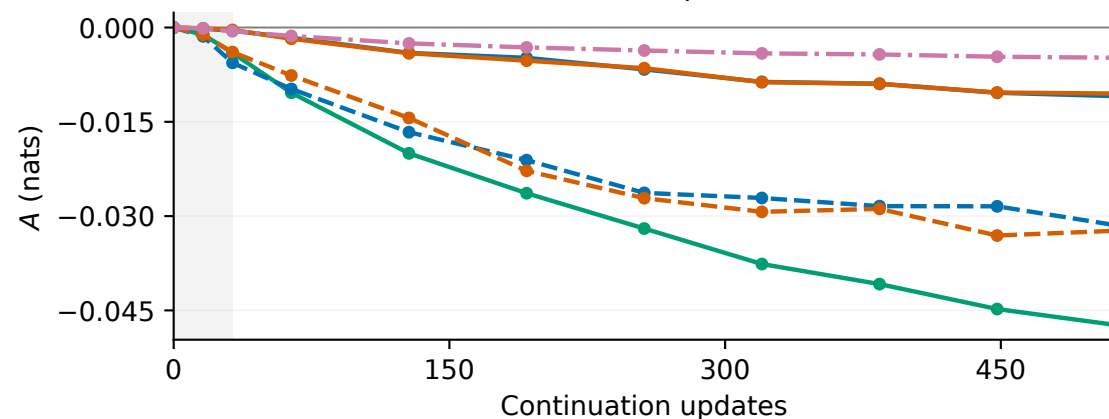


- SGD (smaller rate)
effective $h=3e-06$
- Adam (calibrated)
raw $\eta=2.9e-07$
- Heavy Ball (larger rate)
effective $h=3e-05$
- Heavy Ball (smaller rate)
effective $h=3e-06$
- SGD (larger rate)
effective $h=3e-05$
- Head only
raw $\eta=3e-06$

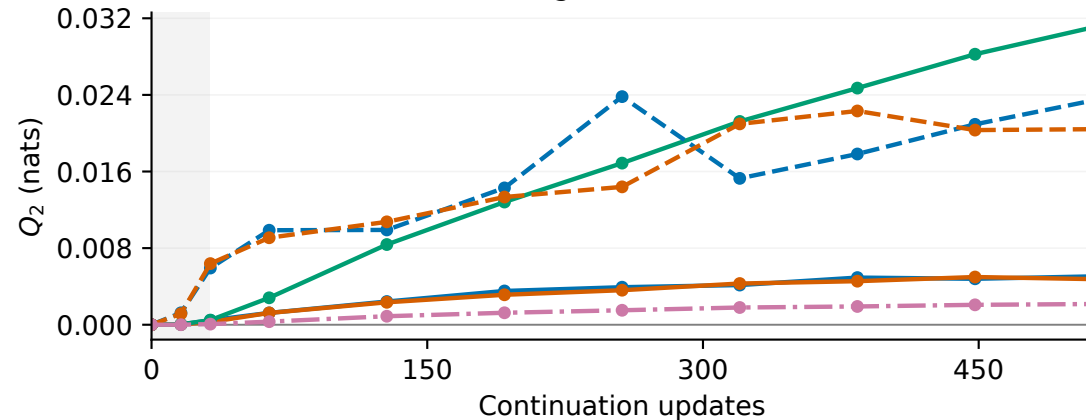
(a) Shared frozen difficulty D



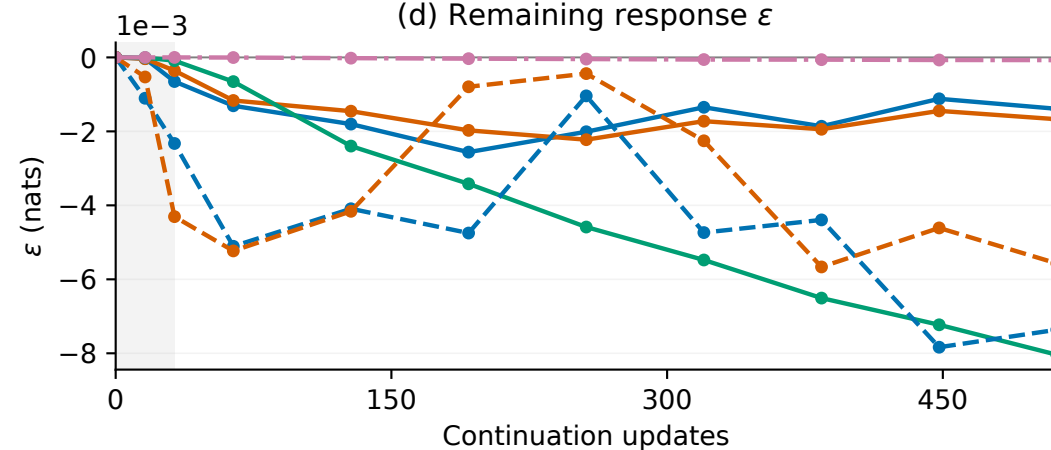
(b) Linear response A



(c) Logit variance Q_2



(d) Remaining response ε



Dots: fixed held-out 32-document readouts; frozen D is horizontal.

Rate-refinement protocol (adaptive); permutation 1 of 3.

Heavy Ball raw $\eta = 0.1h$; Adam rate matches one reset-step Q_2 , not its whole trajectory.

Grey: 32-update warmup. Shared batch order; separate vertical scales; all nats. Rate labels do not imply stability regimes.